SOC Home Lab

Part 2 – Installing Sysmon Configuration on Windows Endpoint

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Technologies Used

- **Oracle VirtualBox**
- **Windows 11 Microsoft Sysmon**
- **SwiftonSecurity**

OBJECTIVE:

The objective of this lab is to Install Microsoft Sysmon and apply SwiftonSecurity configuration to enable enhanced endpoint logging that is not available in native Windows Event Logs.

Lab Setup

Microsoft Sysmon was downloaded from the Microsoft Sysinternals Suite and extracted onto the Windows 11 victim virtual machine. The SwiftonSecurity Sysmon configuration file was then applied during installation to optimize endpoint logging by enabling commonly monitored security events while filtering unnecessary noise.

```
C:\Users\vboxuser\Downloads\Sysmon>sysmon64.exe -accepteula -i sysmonconfig-export.xml


System Monitor v15.21 - System activity monitor
By Mark Russinovich and Thomas Garnier
Copyright (C) 2014-2026 Microsoft Corporation
Using libxml2. libxml2 is Copyright (C) 1998-2012 Daniel Veillard. All Rights Reserved.
Sysinternals - www.sysinternals.com

Loading configuration file with schema version 4.50
Sysmon schema version: 4.91
Configuration file validated.
Sysmon64 installed.
SysmonDrv installed.
Starting SysmonDrv.
SysmonDrv started.
Starting Sysmon64..
Sysmon64 started.
```

Figure 1. Successful installation of Microsoft Sysmon using the SwiftOnSecurity configuration file. The command output confirms that both the Sysmon service and driver were installed successfully on the Windows 11 victim endpoint.

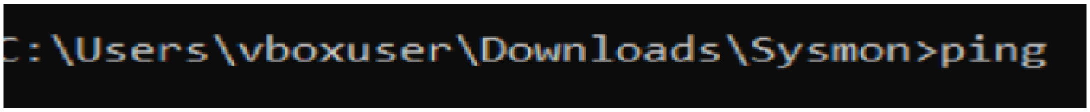

Figure 2. The ping command was executed on the Windows 11 victim endpoint to generate a Process Creation event (Event ID 1) for validating the Sysmon installation.

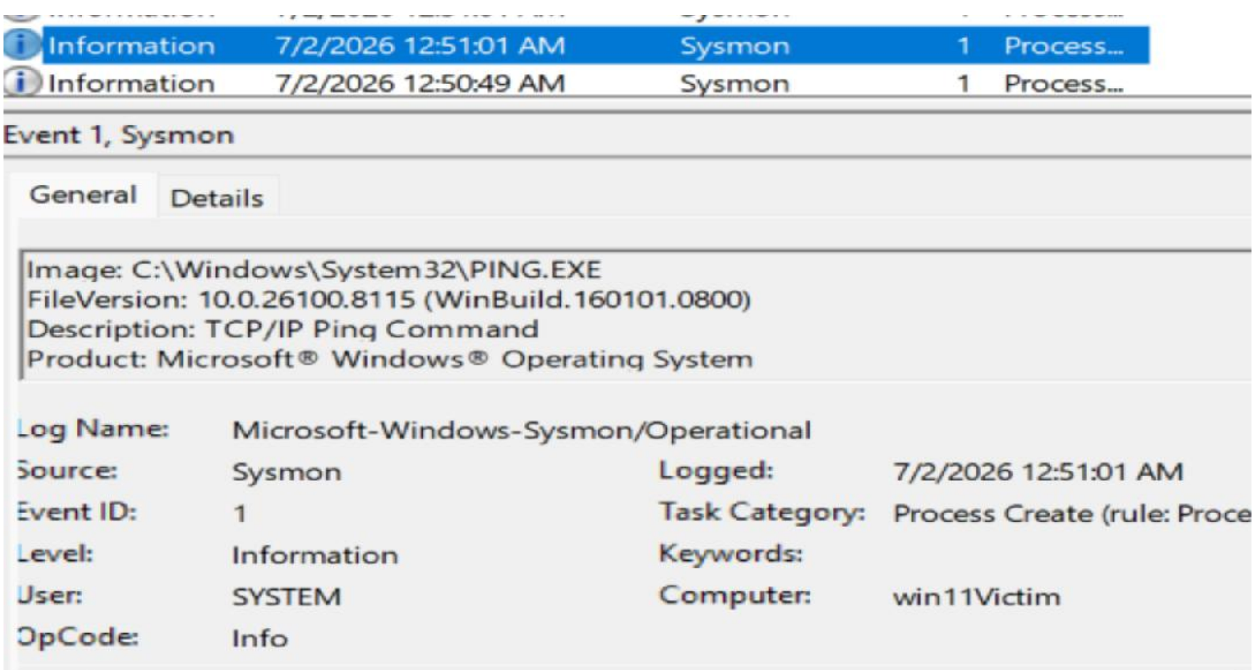

Figure 3. Detailed view of Sysmon Event ID 1 (Process Creation) generated by executing PING.EXE. The event records the executable path, timestamp, user context, and host information, demonstrating the enhanced telemetry provided by Sysmon.

In part 3 I will deploy a separate server VM running Wazuh, an open-source security monitoring platform that provides System Information Event Management (**SIEM**) like capabilities. The Wazuh agent will be installed on the Windows 11 victim VM to forward Windows and Sysmon telemetry to the Wazuh server for centralized log collection, monitoring. and alerting.

Skills Demonstrated

- Oracle VirtualBox
- Windows 11 Virtual Machine Deployment
- Lab Environment Preparation
- Sysmon configuration with SwiftonSecurity
- Endpoint telemetry validation using Sysmon Event ID 1

Key Takeaways

- Successfully configured Sysmon with SwiftonSecurity, providing an extra layer to record and log events that standard Windows Event management can potentially miss.
- Validating Endpoint telemetry with Sysmon Event ID 1, providing details such as the time, execution and location a process was created on system, for monitoring purposes.

Future Labs

Next Steps

- Create and Deploy Wazuh Server
- Install and verify Wazuh agent on Windows 11 victim VM
- Confirm that Sysmon events are successfully ingested and visible within Wazuh